

Example 1

(a) Calculate the thermal noise power available from any resistor at room temperature of 290K for a bandwidth of 1 MHz

$$P_N = K T B \longrightarrow \text{Formula}$$

$$P_N = 1.38 \times 10^{-23} \times 290 \times 1 \times 10^6 \longrightarrow \text{Substitution}$$

$$P_N = 4. \times 10^{-15} \text{ Watts} \longrightarrow \text{Final answer}$$

Example 2

For an electronic device operating at a room temperature of 17°C with a bandwidth of 10 KHz, determine:

- (a) Thermal noise power in watts and in dBm
- (b) RMS noise voltage for 100 ohm internal resistance and a 100 ohm load resistance

$$T = 17^{\circ}\text{C}, B = 10 \times 10^3$$

$$\begin{aligned} T(\text{Kelvin}) &= T^{\circ}\text{C} + 273 \\ &= 17 + 273 \\ &= 290 \end{aligned}$$

$$\begin{aligned} (a) P_N &= K T B \\ &= 1.38 \times 10^{-23} \times 290 \times 10 \times 10^3 \end{aligned}$$

$$P_N = 4 \times 10^{-17} \text{ Watts}$$

Example 2

$$P_N(\text{dBm}) = 10 \log \left(\frac{KTB}{1 \text{ mW}} \right)$$

$$= 10 \log \left(\frac{4 \times 10^{-17}}{1 \times 10^{-3}} \right)$$

$$P_N(\text{dBm}) = -133.97$$

$$\approx \boxed{-134 \text{ dBm}}$$

$$(b) = R = 100 \Omega$$

$$V_N = \sqrt{4 R K T B}$$

$$V_N = \sqrt{4 \times 100 \times 4 \times 10^{-17}}$$

$$V_N = 1.264 \times 10^{-7} \text{ Volts}$$

$$\boxed{V_N = 0.1264 \mu \text{ Volts}}$$

Bring units
in standard form

Example 3 (THD)

- (a) Determine, 2nd, 3rd, and 12th harmonics for a 1 KHz repetitive wave
- (b) Determine: percentage 2nd order, 3rd order and total harmonic distortion for a fundamental frequency with a amplitude of 8Vrms, a second harmonic amplitude of 0.2 Vrms, and third harmonic amplitude of 0.1 Vrms

$f = 1 \text{ KHz}$

(a) 2nd harmonics = $2 \times f = 2 \text{ KHz}$

3rd harmonics = $3 \times f = 3 \text{ KHz}$

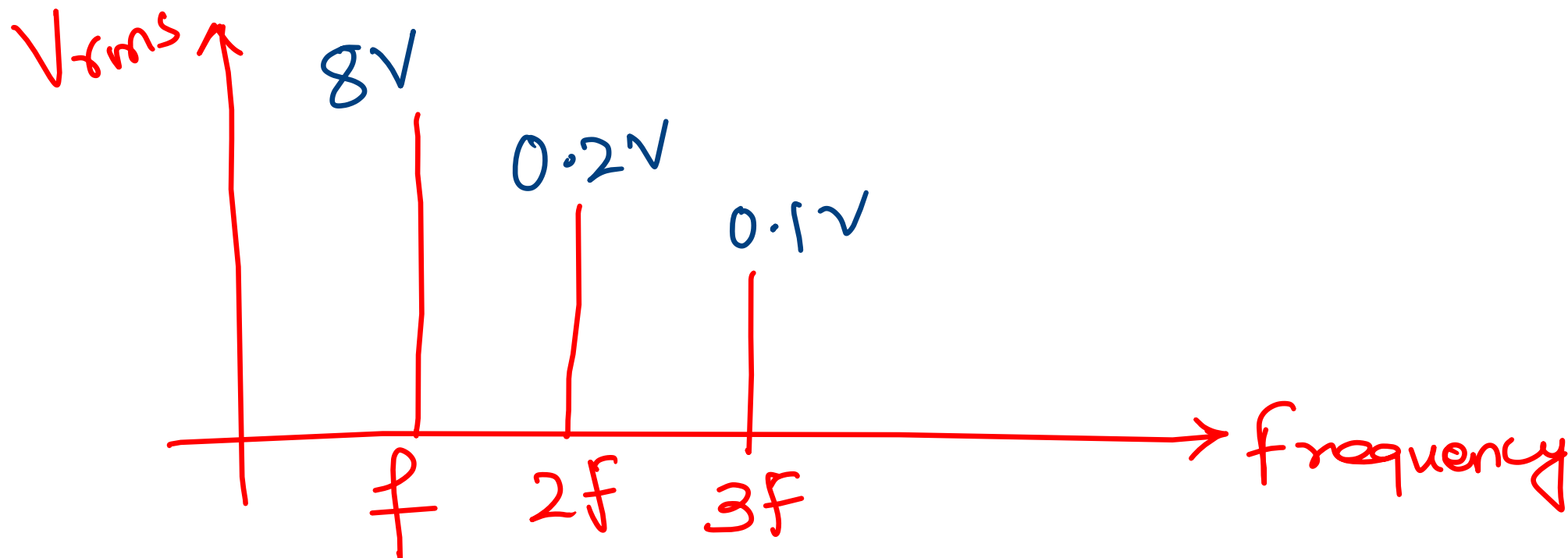
12th harmonics = $12 \times 1 = 12 \text{ KHz}$

$$V_1 = 8 \text{ V}$$

$$V_2 = 0.2 \text{ V}$$

$$V_3 = 0.1 \text{ V}$$

Example 3



Example 3

$$\% 2^{\text{nd}} \text{ order harmonics} = \frac{V_2}{V_1} \times 100 = \frac{0.2}{8} \times 100 = 2.5\%$$

$$\% 3^{\text{rd}} \text{ order harmonics} = \frac{V_3}{V_1} \times 100 = \frac{0.1}{8} \times 100 = 1.25\%$$

$$\text{THD} = \frac{\sqrt{(V_2)^2 + (V_3)^2}}{V_1} \times 100 = \boxed{2.8\%}$$

Example 4

For a nonlinear amplifier with two input frequencies, 3 KHz and 8 KHz, determine:

- (a) First three harmonic present in the output for each input frequency
- (b) Cross product frequencies produced for values of m and n of 1 and 2

When $F = 3 \text{ KHz}$

$$\text{2nd Order Harmonics} = 2 \times F = 2 \times 3 = 6 \text{ KHz}$$

$$\text{3rd Order Harmonics} = 3 \times F = 3 \times 3 = 9 \text{ KHz}$$

$$\text{4th Order Harmonics} = 4 \times F = 4 \times 3 = 12 \text{ KHz}$$

Example 4

$$F_1 = 3 \text{ KHz}$$

$$F_2 = 8 \text{ KHz}$$

$$m = 1, n = 2$$

$$\text{Cross Products} = mF_1 \pm nF_2$$

m	n	Cross Product
1	1	$1 \times 3 \text{ KHz} \pm 1 \times 8 \text{ KHz} = 11 \text{ KHz}, 5 \text{ KHz}$
1	2	$1 \times 3 \text{ KHz} \pm 2 \times 8 \text{ KHz} = 19 \text{ KHz}, 13 \text{ KHz}$
2	1	$14 \text{ KHz}, 2 \text{ KHz}$
2	2	$22 \text{ KHz}, 10 \text{ KHz}$

Example 5

For a non-ideal amplifier following are the parameters. Determine:

(a) Input SNR (dB) (b) Output SNR (dB) (c) Noise Factor (d) Noise Figure

Input signal power: $2 \times 10^{-10} W \rightarrow P_{si}$

Input noise power: $2 \times 10^{-18} W \rightarrow P_{ni}$

Power gain: 1,000,000 $\rightarrow G$

Internal noise: $6 \times 10^{-12} W \rightarrow N$

$$\begin{aligned}
 (a) \text{ Input SNR} &= \frac{P_{si}}{P_{ni}} = \frac{2 \times 10^{-10}}{2 \times 10^{-18}} = 10^8 \text{ Watts} \\
 &= \boxed{80 \text{ dB}}
 \end{aligned}$$



$$P_{So} = \text{Gain} \times P_{Si}$$

$$P_{So} = 200 \mu\text{W}$$

$$P_{no} = \text{Gain} (P_{ni}) + \text{Internal Noise}$$

$$P_{no} = 8 \text{ pW} = 8 \times 10^{-12} \text{ Watts}$$

$$(b) \text{ output SNR} = \frac{P_{So}}{P_{no}} = \frac{200 \times 10^{-6}}{8 \times 10^{-12}} = 2.5 \times 10^6 = 73.98 \approx \boxed{74 \text{ dB}}$$

Example 5

(c) Noise factor (F)

$$F = \frac{\text{Input SNR}}{\text{Output SNR}} \text{ (dB)}$$

$$F = \frac{1 \times 10^8 \text{ W}}{25 \times 10^6 \text{ W}}$$

$$F = 4$$

(d) Noise figure (NF)

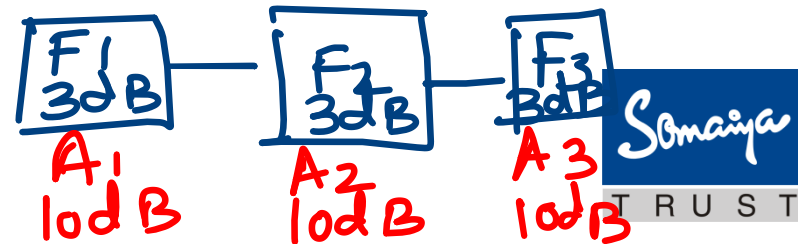
$$\begin{aligned} NF &= 10 \log(F) \\ &= 10 \log(4) \end{aligned}$$

$$NF = 6 \text{ dB}$$

$$\begin{aligned} NF(\text{dB}) &= \text{Input SNR}(\text{dB}) - \text{Output SNR}(\text{dB}) \\ &= 80 \text{ dB} - 74 \text{ dB} \end{aligned}$$

$$NF(\text{dB}) = 6 \text{ dB}$$

Example 6



For three cascaded amplifier stages, each with noise figures of 3dB and power gains of 10 dB, determine the total noise figure

$$F_T = F_1 + \frac{F_2 - 1}{A_1} + \frac{F_3 - 1}{A_1 A_2}$$

$$NF = 10 \log(F)$$

$$F = \text{antilog} \left(\frac{NF}{10} \right)$$

$$F = \text{antilog} \left(\frac{3}{10} \right)$$

$$\boxed{F = 2}$$

$$A_1, A_2, A_3 = \text{antilog} \left(\frac{10}{10} \right)$$

$$\boxed{A_1 = 10}$$

$$F_T = 2 + \frac{2-1}{10} + \frac{2-1}{10 \times 10}$$

$$\boxed{F_T = 2.11}$$

$$NF_T = 10 \log(F_T) = \boxed{3.2 \text{ dB}}$$

Example 7

Determine: (a) Noise figure for an equivalent noise temperature of 75K (Use 290 as reference temperature)

(b) Equivalent noise temperature for a noise figure of 6 dB.

Linear Scale

$$(a) F = 1 + \frac{T_e}{T}$$

$$F = 1 + \frac{75}{290}$$

$$F = 1.258$$

$$NF = 10 \log(1.258) = 1 \text{ dB}$$

$$(b) T_e = T(F - 1)$$

$$F = \text{anti log} \left(\frac{6}{10} \right) = 4$$

$$T_e = 290(4 - 1)$$

$$T_e = 870 \text{ K}$$

Example ~~9~~ 8

- Consider a LNA in a communication system, which amplifies a weak signal. The input signal has an SNR of 20 dB and output signal has an SNR of 15 dB after passing through amplifier. Calculate noise factor and noise figure

$$\text{Input SNR} = 20 \text{ dB}$$

$$\text{output SNR} = 15 \text{ dB}$$

- $\text{SNR (dB)} = 10 \log \text{SNR (linear)}$

- $\text{SNR (linear)} = \text{antilog} (\text{SNR(dB)}/10)$

$$F = \frac{100}{31.62} = 3.16$$

$$\text{NF} = 10 \log (F) = \boxed{5 \text{ dB}}$$

$$(\text{Input SNR})_{\text{dB}} = 10 \log (\text{Input SNR})_{\text{ratio}}$$

$$(\text{SNR})_I = \text{antilog} \left(\frac{20}{10} \right) = 100$$

$$(\text{SNR})_O = \text{antilog} \left(\frac{15}{10} \right) = 31.62$$

Example ~~9~~ 8

- Noise Factor = 3.16 (Linear ratio)
- Noise Figure = 5.015 dB
- Interpretation: System introduces the additional noise that reduces the SNR by about 5 dB.
- Higher the noise figure → worst system performance
- Lower the noise figure → desired (minimal noise while amplifying signal)

Example ~~10~~ 9

Given: $P_{si}=150\mu\text{W}$, $P_{ni}=1.5\mu\text{W}$, $P_{so}=1.5\text{W}$, $P_{no}=40\text{mW}$

Determine Noise factor and noise figure

$$\text{Input SNR} = 100$$

$$\text{Output SNR} = 37.5$$

$$F = 2.67$$

$$\boxed{\text{NF} = 4.2 \text{ dB}}$$

Example 10

SNR at input is 40 dB and noise figure of amplifier is 20 dB, calculate SNR at output.

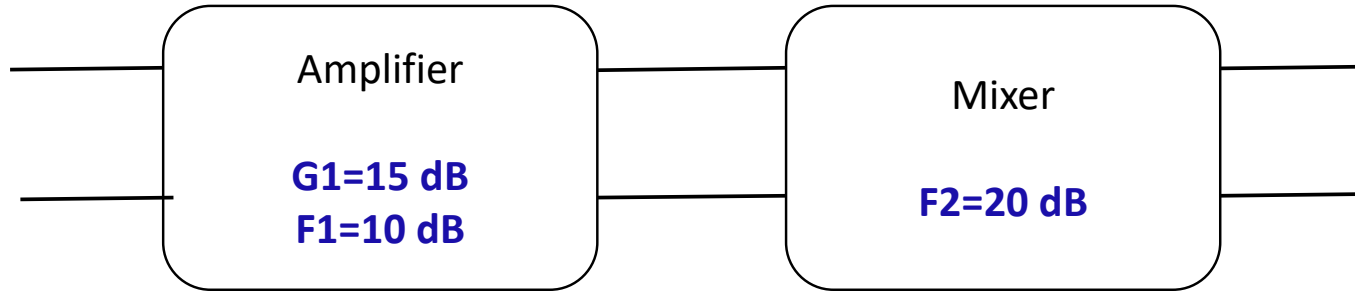
$$NF = (\text{Input SNR})_{dB} - (\text{Output SNR})_{dB}$$

$$(\text{Output SNR})_{dB} = (\text{Input SNR})_{dB} - NF$$

$$= 40 - 20$$

$$= \boxed{20 \text{ dB}}$$

Example ~~12~~ 11



Determine noise figure and noise factor of above circuit

$$F_1 = \text{antilog} \left(\frac{10}{10} \right) = 10$$

$$G_1 = \text{antilog} \left(\frac{15}{10} \right) = 31.62$$

$$F_2 = \text{antilog} \left(\frac{20}{10} \right) = 100$$

$$F_T = F_1 + \frac{F_2 - 1}{G_1} = 13.12$$

$$NF_T = 10 \log(F_T) = \boxed{11.18 \text{ dB}}$$